

Investigations in Learning Analytics

Bridging Gaps in Higher Education through Data Insights

What is Learning Analytics (LA)?

Broadly:

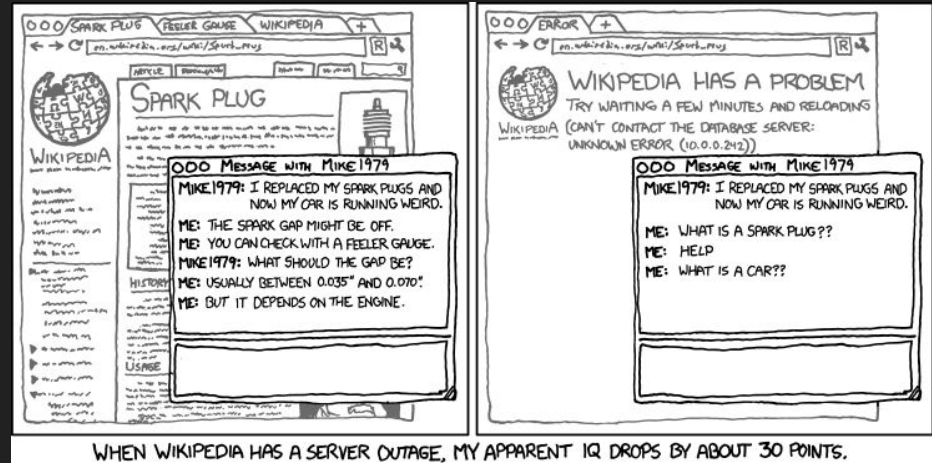
- Measuring, collecting, analysis and reporting of data to garner insights into interactions between learners and the systems they participate in.

What data? In our Institute:

- Assessment data (granular and aggregated)
- Measures of engagement (attendance, assignment completion status, contributions to discussions, on-campus study space visits, use of university digital network and engagements with SLD)
- Measures of self-assessment (reflective tools for LA i.e. focus group data)

Our motivation surrounding the use of LA.

- Transition to higher education often a process spanning several years, with a lot of conceptual schema assimilation > students lack essential self-regulation skills which can be scaffolded with LA
- Aligns with pedagogical framework of student-centered constructivism
- Examining LA through a connectivist approach, leads to interesting thought experiments regarding learning and applicable use cases of LA for the student body.



Research Aims

Three research questions to focus on;

- What insights can we garner from past research and current research with students for an institute-wide LA implementation?
- How can institutes best navigate the complex ethical and legal playing field surrounding analytics?
- What are the major theoretical factors involved in a successful developmental period and implementation of a LA system?

Challenges in the field of LA

- Lack of underlying pedagogical framework
- Lack of rigorous large-scale evidence for effectiveness of LA implementations
- Hot topic regarding the replication crisis
- Highly contextual nature of the data
- Sensitive nature of data often leads to restrictive and selectively funded research
- Or, pseudo-anonymised data that is difficult to interpret holistically

TYPES OF LEARNING ANALYTICS PAPERS



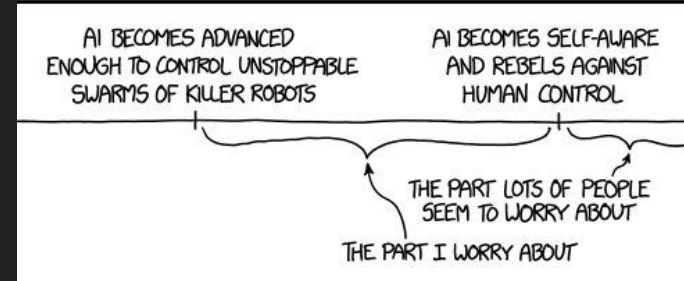
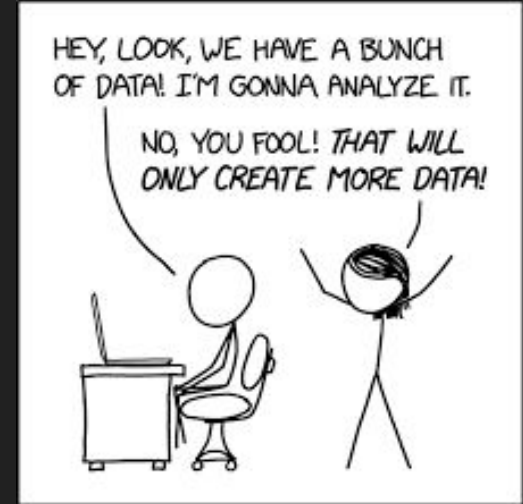
read the original version at <https://xkcd.com/2456/>

Novel Methodology

Lack of in-depth student-centered LA development in prior research, so we created a novel research methodology!

- Use of a prototype as an information-gathering device for insights
- Participants initially conduct one on one interviews to garner individual thoughts and idea
- Develop the prototype as a team

Changes in insight are tracked,



Key Findings

- The novel methodology worked well
- Learning more about LA does seem to diminish fears surrounding LA!
- Students are very concerned around privacy and the commodification of personal data.
- Students prefer humans for the most part.
- Needs are diverse.
- LA is *probably* inevitable



Conclusion

Identified theoretical factors

- Trust in code and the institute, artificial motivation, unique aspects involved in education, questions around fairness, behaviourism, surveillance capitalism, profiling, data autonomy, self-evaluation, and the importance of maintaining human connections despite changes in relationship.

Through the iterative design and evaluation of the prototype, students' perceptions of LA as well as the discussion around the identified theoretical factors shifted positively, demonstrating the potential of participatory design in addressing ethical concerns and enhancing the acceptability of LA systems.

Developing a positive institute culture surrounding LA, particularly if it will be an inevitable aspect of higher education, is critical.

Limitations and Future Work

Limitations:

- It is just me encoding 7.2 hours of transcripts - potential for bias.
- Selective participation group consisting of people that can withstand the chaos of participatory design conducted by me.

Future work:

- Implementing the underlying theoretical machine learning models using a connectivist lens to map out individual knowledge bases that students may have.
- Expanding on the novel methodology to include a diverse range of students.
- Longitudinal study assessing long term impact of an implemented institute-wide LA
- Exploration on evaluating success of an institute-wide LA system that does not reinforce the negative ramifications of Goodhart's Law



Thank You

Thank you for your attention throughout this presentation! All of this work would not have been possible without my supervisor, Alistair Morrison, and his dedication to talking me into staying on scope. I appreciate it!!!

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Image Reference: Munroe, R. (2007) Xkcd. <https://xkcd.com/>

